

Tri-state / bus

A shared bus lets several drivers connect to one wire, but only one should drive at a time

Enable, float, fight

- Starter case: driver A enabled with data one, B and C off, the bus is one
- Turn all enables off with no pull and the bus is Z
- Enable B with zero while A still drives one and you get contention X
- Two drivers on the same value may still show that bit
- Safe protocol: exactly one enable high, often one-hot output-enable in real designs

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Tri-state / bus

INTERACTIVE TOOL

Tri-state / bus contention

Enable one driver at a time on a shared net. **High-Z** floats; two drivers fighting → **contention (X)**.

Starter example: driver A enabled with data 1; B and C off → bus = 1. Enable B with 0 → contention X.

Load starter example

Challenges (1/22)

Quiz: Z. Disconnected / floating level is written? Answer: Z

Your answer

Show hintCheckNextIdle

quiz-zquiz-xquiz-oequiz-onehotstartersee-zfightagree safetoggle-entoggle-datapull-up-zpull-down-zexplainquiz-pullc-drivequiz-fpgathree-fightquiz-impedancercoverdata-ignoredfull-path

Core ideas

High-Z
Enable off — driver disconnected; bus floats.

One driver
Exactly one enable high → clean 0/1.

Contention
Two enables with different data → X (fight).

Tri-state bus starter

learn_digital — tri state bus

Workbook practice

- In the workbook track, fill a small table
- Sketch three tri-state buffers on one bus line with enable pins
- With all enables low and pull-up on, what is the bus?
- Name one pitfall: enabling two drivers at once on an external bus

Pitfalls to watch

- Do not treat multi-drive same value as safe, it still violates one-driver discipline
- Internal FPGA fabrics often prefer muxes over tri-state
- And remember: the browser lab is literacy
- Real boards still need OE timing, bus holders, and protocol rules beyond this toy resolver

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one bus with two drivers and label Z and X cases
- When you are ready, take the short quiz, then continue to barrel shifter